Title Page

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Title. When the Platform Moves Underneath: Maintaining Complementary IT Assets under Endogenous Release Cadence

Running title. Maintaining Complementary IT Assets on Evolving Platforms

Authors and affiliations

1. [First author name], [Department, Institution, City, Country]
2. [Second author name, if any], [Department, Institution, City, Country]

Corresponding author. [Name], [Postal address], [Email: confirm the address to use], [Telephone, optional]

Abstract

Firms increasingly build complementary IT assets, such as fine-tuned adapters, on foundation-model platforms that they do not control, and every platform upgrade destroys some or all of that value. We study this post-adoption maintenance problem as an interaction between the adopter's real-option decision and the platform owner's choice of release cadence and compatibility support. At each upgrade a firm can copy its asset, which is free but imperfect, transfer it with a costly compatibility tool that recovers a share R of its value, or retrain it, an irreversible investment that expires at the next upgrade. The retraining threshold rises with the recovery rate as $1/(1 - R)$ and also with upgrade intensity and value volatility. A free copy weakly dominates immediate retraining, and a costly tool is optimal only for assets of intermediate value and only when its cost falls below a bound Γ. When the platform owner chooses how often to release, an owner that captures part of enterprise surplus over-releases if and only if the average customer is harmed at the margin by a faster clock. Compatibility tools and release speed are strategic complements for any distribution of asset values, so better tools invite faster release and can leave firms that retrain anyway worse off. Under logit duopoly with a release race, each owner internalizes exactly one-third of the marginal change in customer surplus, and numerical analysis shows that an intermediate race separates owners into a fast, compatible frontier platform and a slow, stable one. Release histories of six open-weight model families imply about two main-line versions per year. Recovery measured on open model pairs of 160M to 3B parameters, on three tasks, falls with the size of the upgrade, and a costly transfer method has room to pay in only two of 24 pair-task combinations. The practical rule that follows is to copy by default, to buy a transfer tool only when its measured cost is below Γ, and to read the arrival of owner-provided compatibility tooling as a signal of faster release. Compatibility tooling is thus a governance variable that the platform owner sets jointly with the pace of change.

Keywords. platform evolution; post-adoption maintenance; real options; compatibility; foundation models; release cadence; complementary assets; platform competition

Manuscript details

- Manuscript length: about 6,300 words (main text, excluding references and appendices), 27 pages including references, appendices, tables, and figures.
- Tables: 7 in the main text and 4 in the appendices. Figures: 5.
- Online appendix material: Appendices A to F are included at the end of the manuscript file.
- Replication materials: scripts, per-seed result files, and release-history data (anonymized copy available to reviewers on request).

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